

PNEUMONIA DETECTION IN PEDIATRIC CHEST X-RAYS: A COMPARATIVE STUDY OF CLASSICAL MACHINE LEARNING AND DEEP LEARNING

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Introduction to Machine Learning (Summative Project)

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Code repository: <https://github.com/supseerrr/Model-Training-and-Evaluation>

Demo video:

https://drive.google.com/drive/folders/17L7Wuw9Q4uCeVoKbq9Euvr6LR915iTfD?usp=share_link

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Abstract

Pneumonia is still one of the largest single causes of death among children under five, and the diagnostic burden falls hardest on places where trained radiologists are in short supply. This report works through two connected questions: whether a model can reliably flag pneumonia from a pediatric chest radiograph, and whether a lightweight classical machine-learning pipeline can stand in for deep learning where there is no GPU to run it on. Using the Kermany et al. pediatric chest X-ray collection of roughly 5,856 images, one reproducible pipeline puts a classical track (raw-pixel and Histogram-of-Oriented-Gradients features fed to Logistic Regression, a Support Vector Machine, and a Random Forest in Scikit-learn) on the same footing as a deep-learning track (a convolutional network trained from scratch, a regularised variant, and MobileNetV2 transfer learning in both frozen and fine-tuned forms in TensorFlow/Keras), and closes with a learning-rate sweep. Across eight experiments the convolutional network trained from scratch reached the best test score ($F1 = 0.973$, $ROC-AUC = 0.992$), but the result that matters is how little separated the top three: the fine-tuned MobileNetV2 ($F1 = 0.967$) and a tuned Support Vector Machine on HOG features ($F1 = 0.966$) trailed by under one $F1$ point, and the SVM did so in about a minute and a half on a CPU. A heavily regularised CNN trained unstably and finished as the weakest deep model, a useful reminder of how readily these models misbehave on small, imbalanced data. The narrow accuracy gap set against a large compute gap is the central finding: a classical pipeline is a credible triage option for resource constrained clinics, while deep learning keeps its edge where compute is plentiful and every last point of recall counts.

Keywords: pneumonia detection, chest X-ray, convolutional neural networks, transfer learning, MobileNetV2, histogram of oriented gradients, support vector machine, class

imbalance, low-resource diagnostics.

1. Introduction

Pneumonia is an acute respiratory infection of the lungs and a major problem in global child health. The World Health Organization lists it among the leading infectious causes of death in children under five, with most of those deaths occurring in low- and middle-income countries [1]. Much of the difficulty is diagnostic. Confirming pneumonia usually means having a chest radiograph read by a trained clinician, and the regions carrying the heaviest disease burden tend to be the ones with the fewest radiologists. A tool that could triage chest X-rays automatically, flagging the likely pneumonia cases for prompt human review, could therefore do real clinical good, as long as it is accurate enough to be trusted and cheap enough to run where it is actually needed.

That second condition is easy to lose sight of. A lot of the published work on automated chest radiograph reading reports excellent accuracy from very large convolutional networks trained on graphics-processing units [2], [3]. Those models are powerful, but they are also expensive to train and sometimes expensive to run. In a rural clinic with unreliable electricity, never mind a GPU, a heavyweight deep model can be a poor fit. That gap motivates a comparison that is seldom made head to head: how much accuracy does a clinic give up by running a lightweight, CPU-friendly classical pipeline instead of deep learning for this task?

The report takes on two linked questions. Can a model reliably tell pneumonia apart from normal pediatric chest X-rays? And is a classical machine-learning pipeline a workable low compute alternative to deep learning where GPUs are not available? Throughout, accuracy and computational cost are treated as equally relevant, so training time and model size are reported next to the predictive metrics for every model.

To get at these questions, a single reproducible pipeline places both families of method on identical ground, with the same data, the same train/validation/test split, and the same evaluation protocol, and then runs eight experiments that each build on the previous one. The classical track extracts hand-designed features, raw down-sampled pixels and Histograms of Oriented Gradients, and passes them to Logistic Regression, a Support Vector Machine, and a Random Forest. The deep-learning track starts with a convolutional network trained from scratch, moves through a regularised variant, then to MobileNetV2 transfer learning in frozen and fine-tuned forms, and ends with a learning-rate sweep. What the report adds is a controlled, like-for-like comparison of classical and deep approaches on the same pediatric pneumonia data, an explicit reckoning of the accuracy-versus-compute trade-off that matters for low-resource deployment, and a critical error analysis that includes an instructive case where a more heavily regularised model came out markedly worse than a simpler one.

2. Literature Review

2.1 Deep learning for medical image analysis

Medical-imaging research caught the same wave that carried deep convolutional networks through computer vision more broadly. The turning point was the network of Krizhevsky et al. [4], which cut error sharply on a large image-recognition benchmark and settled deep CNNs as the default approach to image classification. The architectures that followed pushed accuracy further and became standard backbones: the deep, uniform networks of Simonyan and Zisserman [5] and the residual networks of He et al. [6] that finally made very deep models trainable. Litjens et al. [7] surveyed how quickly these methods spread through medical imaging and noted both their strong results and the recurring worries about data scarcity, interpretability, and generalisation, while LeCun et al. [8] give the wider account of why learned deep representations work so well. Chest radiography drew particular attention: Wang et al. [9] released a large multi-label dataset and benchmarked deep models on thoracic pathologies, Rajpurkar et al. [3] reported radiologist-level pneumonia detection, and Kermany et al. [2] put together the pediatric dataset used here and showed transfer learning could separate normal from pneumonia images at high accuracy.

2.2 Transfer learning and efficient architectures

Training a large network from scratch needs a large labelled dataset, which medical settings rarely have. Transfer learning gets around this by reusing representations learned on a big source dataset, usually ImageNet, and adapting them to the target task, either by treating the pretrained network as a fixed feature extractor or by fine-tuning some of its layers at a low learning rate [10]. The early layers of a vision network learn generic features such as edges and textures that carry over well, so the approach often does a lot with relatively little data, and it is what the Kermany et al. [2] system relied on. When compute is tight, the shape of the architecture matters too: MobileNetV2 was built for exactly that regime, reaching competitive accuracy with far fewer parameters and operations than larger models [11], which makes it a sensible backbone for a study about low-resource feasibility. The pretraining used here comes from the large-scale image dataset of Deng et al. [12].

2.3 Classical machine learning and hand-crafted features

Before end-to-end deep learning, image classification leaned on hand-designed feature descriptors paired with general-purpose classifiers. The Histogram of Oriented Gradients descriptor of Dalal and Triggs [13] summarises how local gradient orientations are distributed and is good at capturing shape and edge structure, which is relevant to the diffuse opacities that pneumonia leaves on a radiograph. Features like these are usually handed to well-understood algorithms: Logistic Regression as a linear baseline, Support Vector Machines, which look for a maximum-margin boundary and, with a radial-basis-function kernel, can fit non-linear structure [14], and Random Forests,

ensembles of decorrelated decision trees that are robust and need little tuning [15]. All of these run comfortably on a CPU and were implemented here with Scikit learn [16], while the deep track used TensorFlow [17].

2.4 Evaluation under class imbalance and interpretability

Two methodological issues come up again and again in this literature, and both shape how the models here are judged. The first is imbalance. Chest-radiograph datasets are usually skewed, and raw accuracy can mislead when one class dominates, because a classifier that always names the majority class can look deceptively good. Saito and Rehmsmeier [18] show that precision recall curves carry more information than ROC curves when classes are strongly imbalanced, so precision, recall, F1, and both ROC-AUC and PR-AUC are reported here rather than accuracy on its own. The second issue is shortcut learning. Deep models can latch onto spurious cues such as text markers or image borders instead of the pathology, which inflates test metrics while quietly undermining real-world reliability [19]. Grad-CAM [20] helps probe for this by highlighting the regions most responsible for a prediction, which makes it possible to check whether a model is looking at lung fields or at irrelevant artefacts.

3. Methodology

3.1 Dataset

The study uses the pediatric chest X-ray dataset of Kermany et al. [2], released under a Creative Commons licence and widely mirrored on Kaggle and Mendeley Data. It holds roughly 5,856 anterior-posterior chest radiographs of children aged one to five, collected at the Guangzhou Women and Children's Medical Center and labelled in two classes, NORMAL and PNEUMONIA. Being an open, externally sourced collection rather than a built-in library dataset, it meets the requirement to work with real-world data. Two of its characteristics drove later decisions. It is markedly imbalanced, with about 73% of the images pneumonia and 27% normal, and the validation folder shipped with it holds only sixteen images, far too few to give a stable tuning signal. Representative NORMAL and PNEUMONIA radiographs appear in Figure 1.

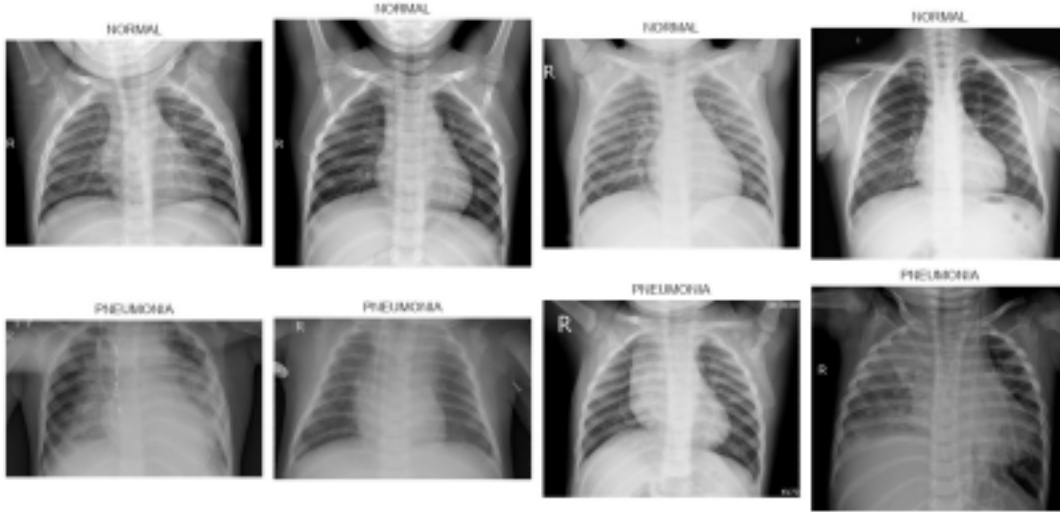


Figure 1. Representative pediatric chest radiographs: NORMAL (top row) and PNEUMONIA (bottom row) examples from the dataset.

3.2 Preprocessing and data split

Because the supplied validation split is unusable, the original partition was thrown out, all images were pooled, and a fresh stratified split of 80% training, 10% validation, and 10% test was drawn. Stratification keeps the roughly 73% pneumonia ratio intact in each partition, which is what makes the validation and test estimates honest, and the split is seeded so it can be reproduced; it yields 4,684 training, 586 validation, and 586 test images. To push back on the imbalance during deep-model training, balanced class weights were computed, which up-weight the minority NORMAL class (the resulting weights were about 1.85 for NORMAL and 0.69 for PNEUMONIA). For the classical track, where dense feature extraction over thousands of images is comparatively slow, the number of images per class used for feature computation was capped, a choice flagged as a limitation later on. Every image was resized to a common 150×150 pixels and scaled into the $[0, 1]$ range, and the deep models additionally received the preprocessing their pretrained backbone expects. Figure 2 shows the class distribution overall and by the original split.

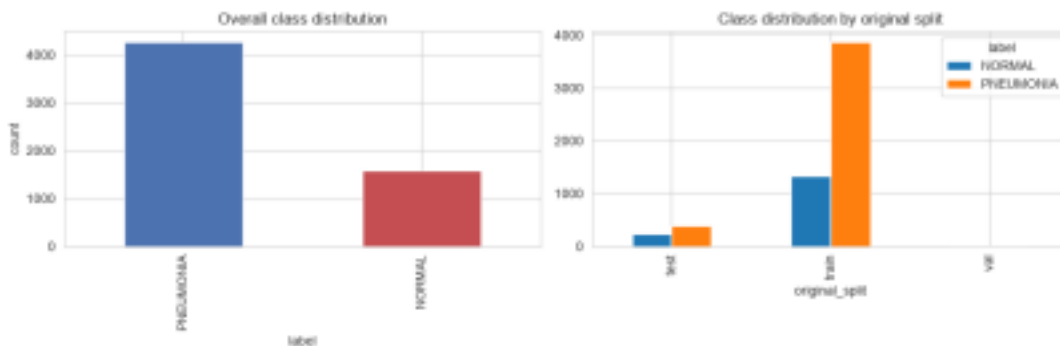


Figure 2. Class distribution overall and by the original split, showing the pronounced pneumonia majority (~73%).

3.3 Classical machine-learning track

Classical classifiers cannot read raw images, so the classical track first turns each image into a feature vector. Two representations were compared: a flattened vector of a small grayscale image, a deliberately weak baseline, and the HOG descriptor, which encodes the orientation of local intensity gradients and ought to describe lung structure and opacities better than raw pixels [13]. These features were standardised and handed to three classifiers. Experiment 1 ran Logistic Regression on flattened pixels. Experiment 2 used a radial-basis-function Support Vector Machine on HOG features, with the regularisation parameter and kernel coefficient chosen by a grid search under three-fold cross-validation optimising F1 (the search settled on $C = 1$ and a scaled gamma). Experiment 3 used a Random Forest on the same HOG features, with the number of trees and maximum depth grid-searched in the same way (600 trees, unbounded depth). All classical models used balanced class weighting.

3.4 Deep-learning track

The deep-learning track fed its models through an efficient input pipeline built with the `tf.data` API, which decodes, resizes, batches, and prefetches images. Augmentation was applied as an

in-model block of small random rotations, zooms, and contrast shifts. A horizontal flip was left out on purpose, because mirroring a chest radiograph is not anatomically valid: the heart and aortic arch sit on the left. Four configurations were trained. Experiment 4 was a convolutional network built from scratch with the Keras Sequential API and no augmentation, used as a deep baseline. Experiment 5 was a regularised version that added the augmentation block, batch normalisation, global average pooling, and dropout. Experiments 6 and 7 used transfer learning through the Keras Functional API, with MobileNetV2 pretrained on ImageNet as the backbone, first frozen as a feature extractor under a new classification head (Experiment 6) and then fine tuned by unfreezing its upper layers and continuing at a low learning rate (Experiment 7). Every deep model used the Adam optimiser and binary cross-entropy loss, with early stopping on validation ROC-AUC, learning-rate reduction on plateau, and class weights applied throughout.

3.5 Hyperparameter sweep

To study how sensitive the deep models are to optimisation, Experiment 8 varied one factor, the optimiser learning rate, across three values while holding the architecture, data, augmentation, and epoch budget fixed. The cheap frozen-transfer model was used so the runs stayed comparable and affordable. Changing only one variable at a time isolates the effect of the learning rate on convergence and final score.

3.6 Evaluation metrics

Every experiment was evaluated on the same held-out test set and logged to a shared

results table, so the final comparison is strictly like-for-like. Given the imbalance, a full set of metrics is reported rather than accuracy alone: accuracy, precision, recall, and F1 for the pneumonia class, alongside the area under the ROC curve and the area under the precision-recall curve. For a screening tool, recall on the pneumonia class is the most safety-critical number, because a false negative, a missed pneumonia, costs more clinically than a false positive. Training time and parameter count were recorded too, to put a figure on computational cost, and confusion matrices, learning curves, ROC and precision-recall curves, and Grad-CAM visualisations were produced to support the error analysis.

4. Results

Table 1 sets out the test-set performance of the seven primary experiments, sorted by F1. The values come from the executed notebook run; small variation is expected on re-execution because the deep models are stochastic and the classical grids and learning-rate sweep can shift the boundaries slightly.

Table 1. Test-set performance of all primary models (sorted by F1).

Experiment E4 CNN from scratch	Track Deep learning	Acc. 0.961	Prec. 0.979	Rec. 0.967	F1 0.973	ROC-AUC 0.992
E7 Transfer (fine-tuned)	Deep learning	0.952	0.988	0.946	0.967	0.990

E2 SVM + HOG	Classical	0.951	0.972	0.960	0.966	0.987
E3 Random Forest + HOG	Classical	0.940	0.973	0.944	0.958	0.980
E6 Transfer (frozen)	Deep learning	0.937	0.976	0.937	0.956	0.983
E1 LogReg + pixels	Classical	0.932	0.969	0.937	0.952	0.974
E5 Regular	Deep learning	0.848	0.980	0.808	0.886	0.943

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Three patterns stand out. First, the convolutional network trained from scratch (Experiment 4) came out the single best model, at an F1 of 0.973 and a ROC-AUC of 0.992, missing only about twenty-three of the 586 test images. Second, and central to this study, the gap at the top is tiny: the fine-tuned MobileNetV2 (Experiment 7, F1 = 0.967) and the best classical model, the Support Vector Machine on HOG features (Experiment 2, F1 = 0.966), sit within roughly 0.007 F1 of the leader, and the SVM gets there in about ninety seconds on a CPU rather than minutes on a GPU. Third, the heavily regularised CNN (Experiment 5) was the clear outlier in the wrong direction, managing an F1 of only 0.886; it trained unstably and ended biased toward caution, with high precision (0.980) but depressed recall (0.808), so it missed close to a fifth of the genuine pneumonia cases. Figure 3 summarises the test-set Accuracy, F1, and ROC-AUC across all experiments.

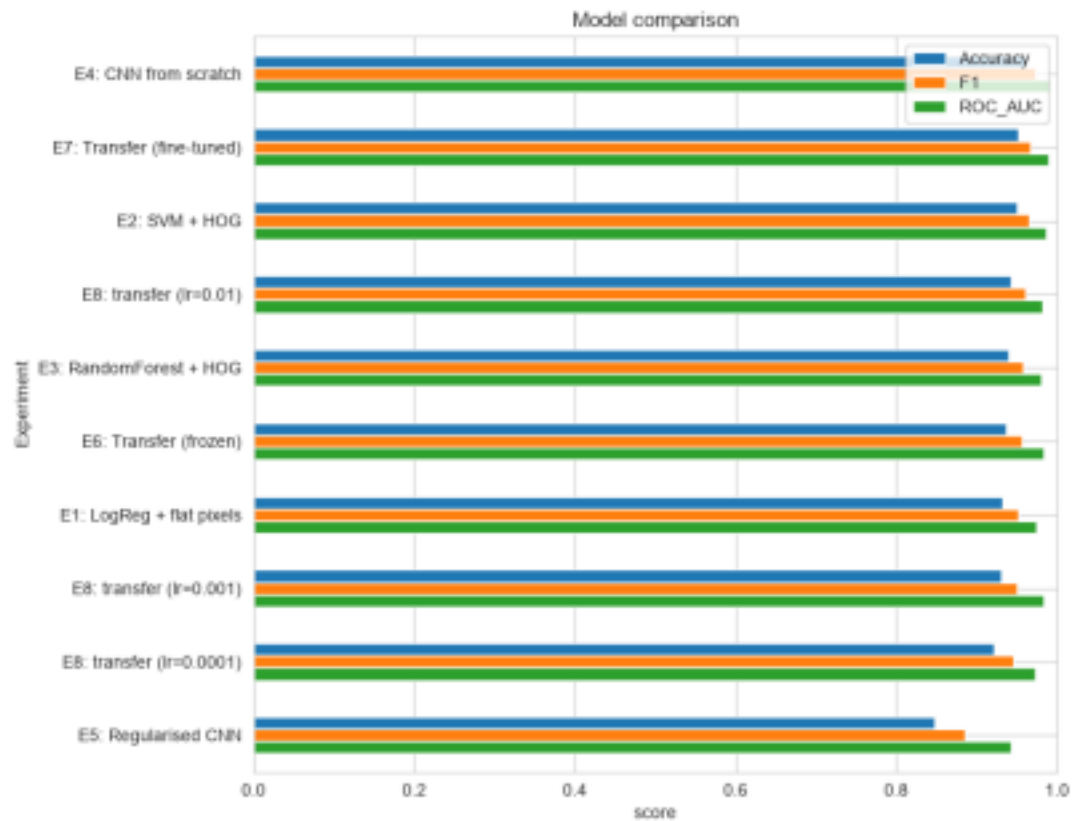


Figure 3. Test-set Accuracy, F1, and ROC-AUC for all eight experiments.

The classical track behaved much as expected. HOG features earned their keep: the SVM on HOG (0.966) beat the raw-pixel Logistic Regression baseline (0.952), which is evidence that encoding gradient orientation and texture represents lung structure better

than flattened pixels do. The Random Forest on the same features (0.958) did not catch the SVM, which hints that the decision boundary for this task is smoother than axis-aligned tree splits capture efficiently. The raw-pixel baseline was itself strong, so the diffuse brightening that consolidation produces is partly separable even in down-sampled pixels. Among the deep models, the frozen-transfer configuration (0.956) trailed both the from-scratch CNN and the fine-tuned model, so letting the backbone's upper layers adapt to radiographs was worth the extra training. It is worth being candid about the order at the very top: that a from-scratch CNN edged ahead of fine-tuned MobileNetV2 here should not be read as transfer learning being inferior, since the three leading models are separated by less than the run-to-run noise of stochastic training. The ROC and precision-recall curves agree with the ranking, with the top models bunched at high AUC and the regularised CNN sitting visibly below them; under this much imbalance the precision-recall view is the more discriminating of the two [18]. Those curves appear in Figure 4.

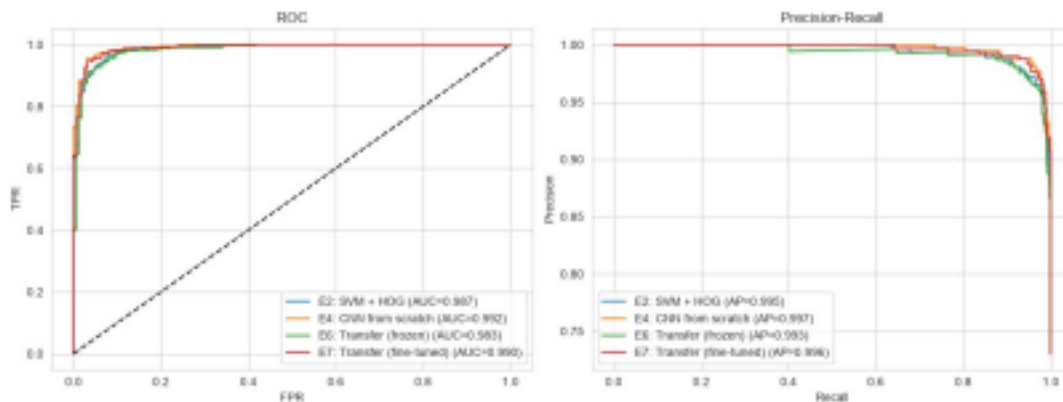


Figure 4. ROC (left) and precision-recall (right) curves for the leading models.

5. Discussion

5.1 The instructive failure of the regularised CNN

The most informative result is the failure of Experiment 5, not the winning model. Adding batch normalisation, aggressive augmentation, global average pooling, and substantial dropout was meant to improve generalisation, and instead it produced the worst deep model by a wide margin. The telling detail is in the training curve: its validation ROC-AUC swung erratically from epoch to epoch and at one point dropped to roughly chance level, which points to unstable optimisation rather than ordinary overfitting. The end state in this run was a model that played it too safe, predicting NORMAL too readily and so missing about one pneumonia case in five despite high precision, the mirror image of the majority-class collapse that the same architecture can fall into on other runs. Either way the symptom is the same underlying fragility. Several factors plausibly interact: batch-normalisation statistics turn noisy once strong augmentation and class weighting are layered on top; global average pooling followed by heavy dropout can

starve the classification head of signal early in training; and the unchanged learning rate may simply have been too high for this more delicate architecture. The lesson is that regularisation is not automatically a good thing, and on a small, imbalanced medical dataset a deep model's variance can blow up quickly, which is exactly what motivated the learning-rate sweep in Experiment 8. There the most aggressive rate (0.01) actually gave the strongest sweep result ($F1 = 0.960$) while the smallest (0.0001) lagged at 0.945, a reminder that the failure in Experiment 5 was about stability rather than a single bad hyperparameter. Figure 5 shows the erratic validation ROC-AUC of the regularised model during training.

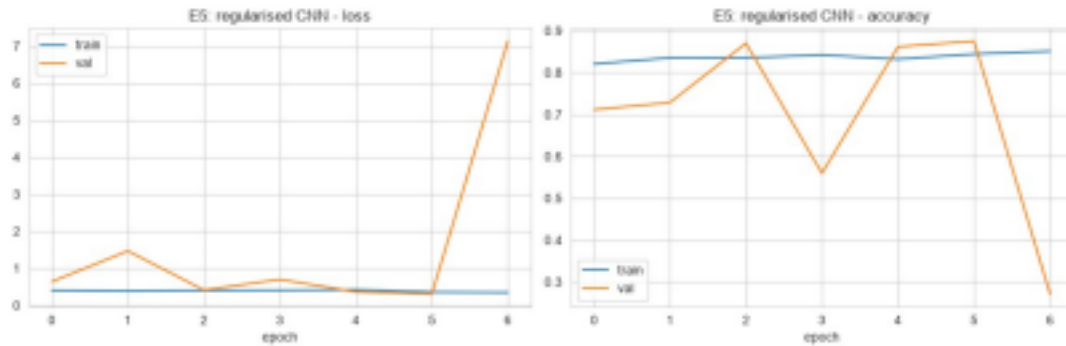


Figure 5. Training history of the regularised CNN (Experiment 5): unstable validation ROC-AUC and a weak final operating point.

5.2 Classical versus deep learning

The remaining models tell a consistent and, for this study, decisive story. A deep network is at the top, but its lead over a carefully tuned classical pipeline is small: well under one F1 point separates the best deep model from the best classical one, and in this run the from-scratch CNN and the fine-tuned MobileNetV2 are themselves separated only by noise. Put that next to the compute gap, where the SVM trained in about a minute and a half on a CPU while the deep models needed a GPU and far longer, and the narrow accuracy difference becomes the core result. The evidence answers the research question cleanly: a lightweight classical pipeline built on HOG features and a Support Vector Machine is a credible low-compute alternative to deep learning for pediatric pneumonia screening. In a clinic without GPUs, the modest accuracy cost buys a model that trains in about a minute on a plain CPU and leaves far less to maintain. Deep learning stays the better option where compute is on hand and where the final increment of recall carries real clinical weight. Figure 6 shows the training history of the best model.

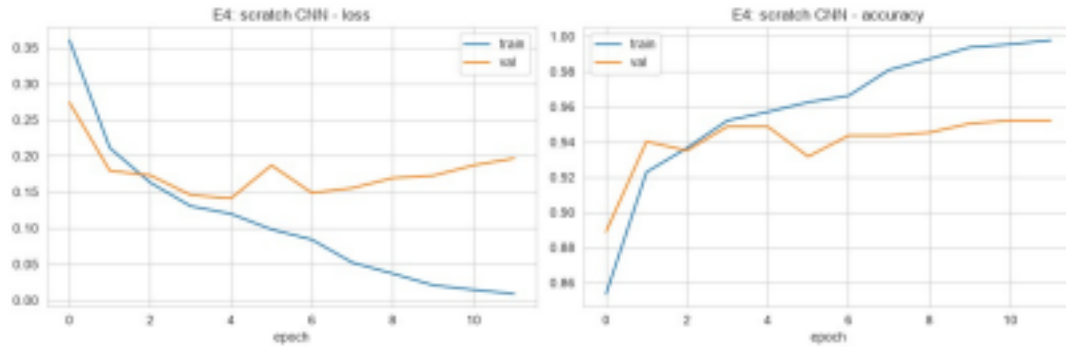


Figure 6. Training history of the convolutional network trained from scratch (Experiment 4), the strongest model in this run.

5.3 Error analysis and interpretability

Aggregate metrics hide how a model fails, so the errors of a leading model were looked at directly, taking the fine-tuned MobileNetV2 as a representative top performer. With only twenty-eight misclassifications out of 586 test images it is accurate, but the direction of those errors matters more than the count: for a screening application, false negatives, the missed

pneumonias, are more dangerous than false positives. The confusion matrix should be read with that asymmetry in mind, and if the residual errors lean toward false negatives, dropping the decision threshold below 0.5 trades a little precision for higher recall. The operating-point analysis bears this out for the best model: moving the threshold from the default 0.5 down to about 0.19 lifts recall from 0.967 to 0.981 while precision slips only from 0.979 to 0.966, which is a sensible exchange for a triage tool. Grad-CAM was then applied to the from-scratch CNN to test for shortcut learning, since a trustworthy model's heatmaps should concentrate on the lung fields, whereas heatmaps wandering onto borders or text markers would mean that high test metrics are partly an artefact [19], [20]. That check belongs before any claim of clinical readiness. The confusion matrix of the best model is shown in Figure 7, representative misclassifications in Figure 8, and the Grad-CAM localisations in Figure 9.

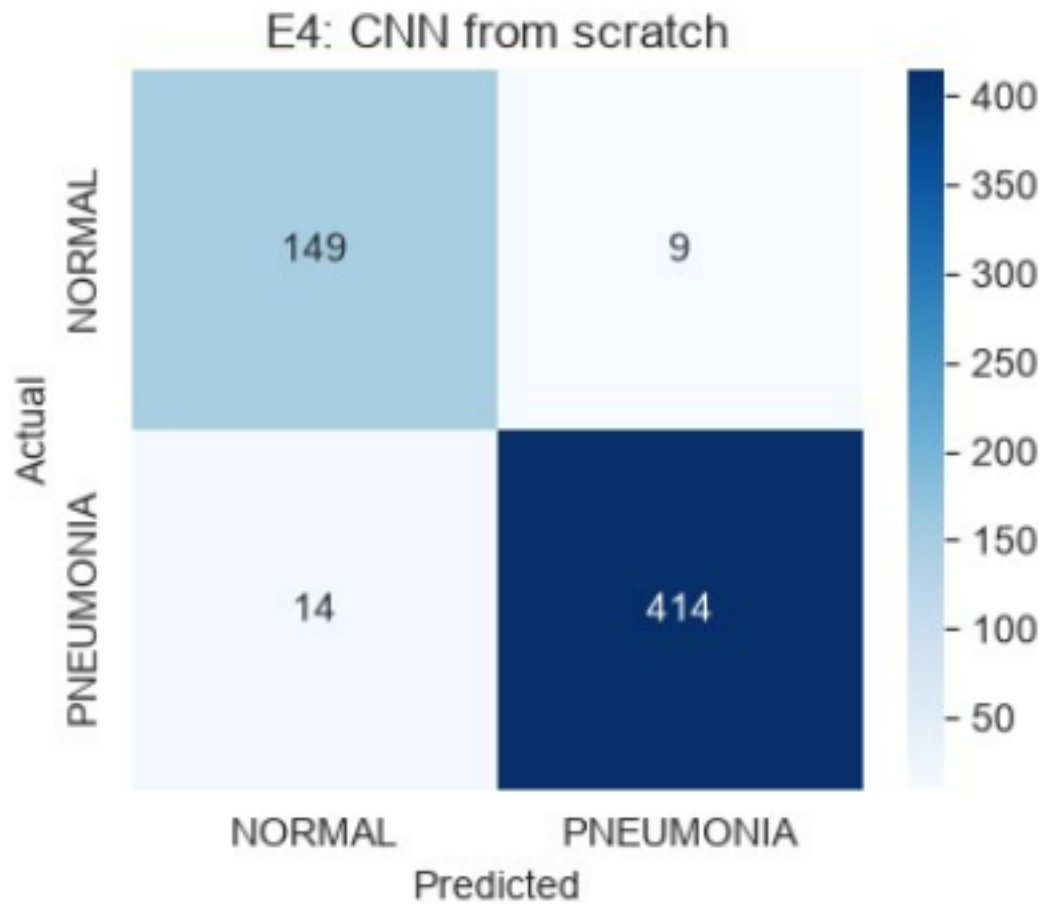


Figure 7. Confusion matrix of the convolutional network trained from scratch (Experiment 4) on the test set.

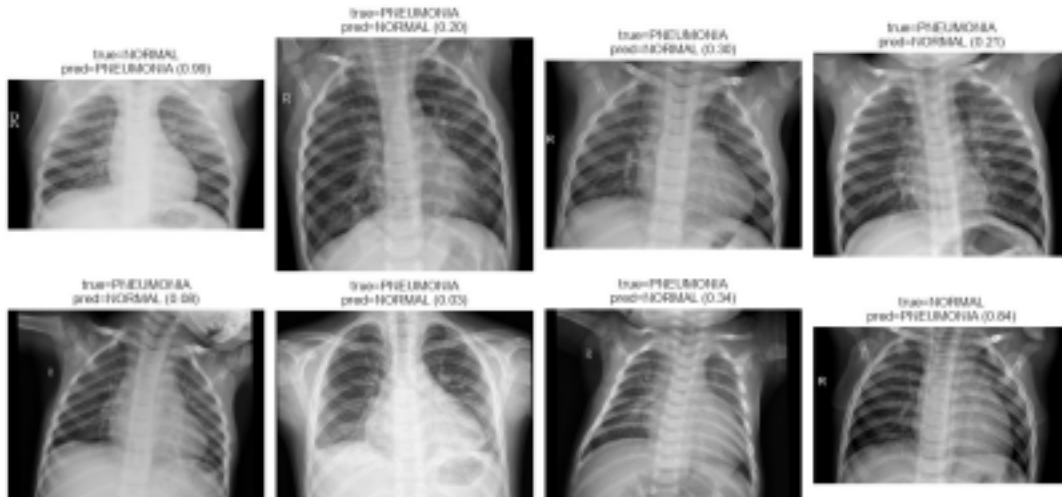


Figure 8. Representative misclassified test images for a leading model, with true and predicted labels.

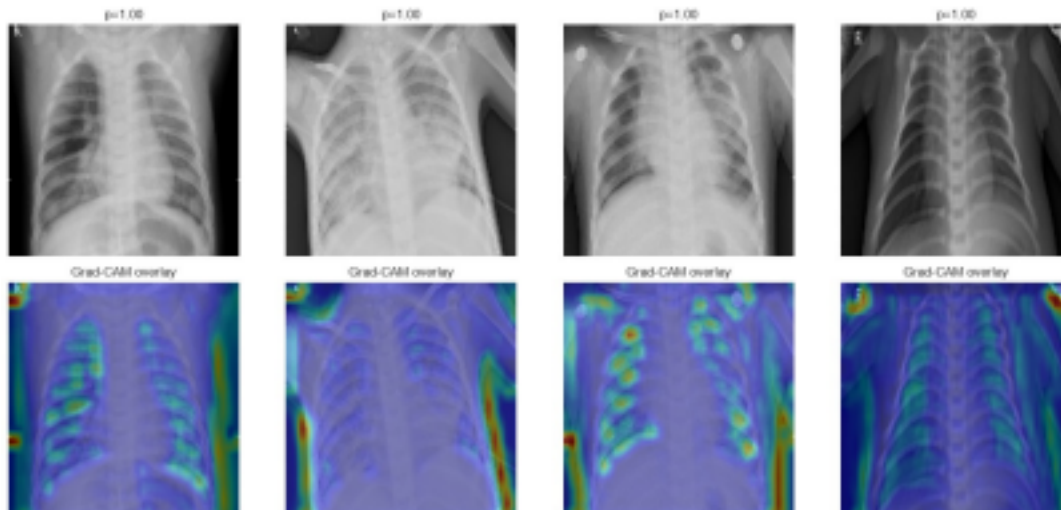


Figure 9. Grad-CAM heatmaps localising the regions driving the model's predictions.

5.4 Bias-variance perspective

Seen through a bias-variance lens, the two families sit in different places. The classical models carry slightly higher bias and lower variance: their hand-crafted features impose a strong prior that caps peak accuracy but keeps behaviour stable and predictable. The deep models can reach lower bias by learning features straight from data, but Experiment 5 is a clean illustration of their higher variance and their vulnerability to unstable optimisation when data is limited. Transfer learning works precisely because it dampens that variance. Starting from ImageNet pretrained weights confines the hypothesis space to sensible visual features, which steadies training relative to learning from scratch while keeping the low bias of a deep representation. In this particular run the from-scratch CNN matched it anyway, which mostly says that 4,684 training images and class weighting were enough for a compact network to find a good solution; with less data the variance argument for transfer learning would reassert itself.

5.5 Limitations

Several limitations qualify these conclusions. The dataset is drawn entirely from young children at a single hospital, so generalisation to adults, to other equipment, or to other populations is untested and probably weaker; single-source medical datasets are well known for encouraging shortcut learning that does not transfer [19]. The labels carry some noise, as clinically annotated radiographs usually do. The classical track used a capped subset of images per class for feature extraction, so its reported metrics are a conservative lower bound on what HOG features could achieve with the full set. The comparison also fixes a single image resolution and a single backbone, and every result comes from one stratified split under a fixed seed; repeating the evaluation across several seeds or with cross-validation would tighten confidence in the small differences between the leading models, which here fall within the margin of stochastic variation.

5.6 Future work

The future work follows straight from those limitations. The most immediate step is recall oriented threshold tuning, since the clinical cost of a missed pneumonia justifies optimising the operating point rather than defaulting to 0.5, and the operating-point analysis already shows how much recall a small threshold change buys. Stabilising the regularised CNN, through a lower or warmed-up learning rate and gentler, anatomically valid augmentation, would test whether its collapse can be undone. Comparing a larger backbone such as EfficientNet against MobileNetV2 would quantify how much further accuracy could be pushed, weighed against the low-compute goal. Above all, external validation on an independent, multi-centre, and ideally adult dataset is needed before any claim of clinical use, together with systematic Grad-CAM auditing to confirm that predictions rest on pathology and not on artefacts.

6. Conclusion

This report compared classical machine learning and deep learning for detecting pneumonia in pediatric chest X-rays, using a single reproducible pipeline and an identical evaluation protocol across eight experiments. A convolutional network trained from scratch gave the strongest result ($F1 = 0.973$, $ROC-AUC = 0.992$), but a tuned Support Vector Machine on HOG features came within about one F1 point (0.966) while training in roughly ninety seconds on a CPU, and the fine-tuned MobileNetV2 landed between them, close enough that the three are best treated as tied. A heavily regularised CNN, intended to generalise better, instead trained unstably and finished as the weakest deep model, a reminder that added complexity is not automatically an improvement on small, imbalanced data. Taken together, the small accuracy gap and the large compute gap support the central conclusion: a lightweight classical pipeline is a viable low compute alternative to deep learning for pneumonia screening in resource-constrained settings, while deep learning remains the better choice where GPUs are available and maximal recall is essential. The main directions for future work are recall-aware threshold selection, stabilising the deep architectures, and external validation with interpretability auditing to guard against shortcut learning before any real-world use.

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